VPMP Polytechnic, Gandhinagar

Department of Computer Engineering – Semester 6

IOT (4360703) - ASSIGNMENTS

Assignment 1

Unit – I (Introduction to IoT)

1. Explain definition and Characteristics of IoT.
2. List out Applications of IOT.
3. Explain Key components of IOT.
4. Explain Architecture of IOT.
5. Explain the challenges involved in developing an IoT system.

Assignment 2

Unit – II (Sensors & Actuators)

1. Define Sensors, Actuators, and Transducers.
2. List types of Sensors and explain any one.
3. List types of Actuators and explain any one.
4. Write a difference between Sensor and Actuator.
5. Explain the need of ADC circuit in analog sensors.
6. Explain need of Relay while using Actuators.

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Assignment 3

Unit – III (Programming with Arduino UNO)

1. Explain Arduino Uno board block diagram with figure.
2. Explain data types in Arduino.
3. List out operators of Arduino and explain any one with example.
4. Explain library functions in Arduino.
5. Demonstrate how to define user defined function in Arduino.
6. Explain I/O functions with an example.
7. Explain char functions with an example.
8. Explain Math functions with an example.
9. Write a program for LED blinking using Arduino.
10. List out serial communication functions and explain any three of them.

Assignment 4

Unit – IV (IoT Communication Protocols)

1. List out Messaging protocols. Discuss MQTT protocol in Detail.
2. Explain Constrained Application Protocol (COAP) in detail.
3. Differentiate CoAP and MQTT.
4. Explain XMPP in detail.
5. Describe Transport layer protocol in detail.
6. Define Topology. List out all topologies in sensor networks and explain any one in detail.

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Assignment 5

Unit – V (Applications of IoT)

1. Explain smart Home Automation system with diagram.
2. Explain Agricultural system with diagram.
3. Explain smart Parking system with diagram.

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